CSE471 | FALL 2026
Assignment 2

Behavioural modelling with UML: use case, activity and sequence diagrams

Part A · Use case diagrams

Questions 1–5. Show the system boundary, every actor and use case, and the relationships between them.

Question 1 MetroPass smart ticketing

MetroPass is a smart-card and mobile ticketing system for a city metro rail. Any visitor can view the route map, station facilities and the fare between two stations without signing in. To buy anything, a passenger must create an account and log in, and the system verifies the passenger's mobile number with a one-time password (OTP) whenever a new device is used.

There are two kinds of passengers: regular passengers and student passengers. Student passengers get a concession fare, which requires their institution ID to be verified by the university database once every semester. Passengers can recharge their card balance by mobile financial service or debit card; every recharge is processed by an external payment gateway. A passenger can also buy a single-journey QR ticket, which follows the same steps as a card recharge but is valid only for that day.

At the gates, the gate reader validates the card at entry and deducts the fare at exit. If the balance is not enough at exit, the passenger may top up at the counter before leaving. Passengers can report a lost card and, if they wish, transfer the remaining balance to a new card. Station officers can top up cards at the counter and resolve gate complaints. The operations manager can change fares and view daily ridership reports, which the system prepares by compiling data from every gate.

Design a use case diagram based on the above scenario.

Question 2 CareLink outpatient services

CareLink is an outpatient service platform for a private hospital in Chattogram. Anyone can search for doctors by department or name and see their weekly schedules. Patients must register and log in to book an appointment, and while booking, the system always checks whether the chosen doctor still has a free slot. If a patient marks the visit as an emergency, the patient may also request a priority slot, which must be approved by the duty manager.

Patients can pay the consultation fee online through a payment gateway or pay at the hospital counter. Returning patients can view their prescriptions and laboratory reports. Doctors are either consultants or medical officers. Both can view their daily appointment list and write e-prescriptions, but only consultants can refer a patient for surgery. Whenever a doctor writes an e-prescription, the medicine list is sent to the hospital pharmacy system.

Laboratory technicians upload test reports, and the patient is notified by SMS once a report is uploaded. The front desk officer can reschedule or cancel appointments when a doctor is unavailable; when an appointment is cancelled, any online payment is refunded automatically. The hospital administrator manages doctor profiles and department information.

Design a use case diagram based on the above scenario.

Question 3 KrishiBazaar crop marketplace

KrishiBazaar is an online marketplace that connects farmers directly with wholesale buyers across Bangladesh. Visitors can browse crop listings and today's market prices. Farmers and buyers must register with their NID, and registration is completed only after the NID is verified through the government NID server.

A farmer can post a crop listing with the quantity, expected price and harvest date, and the system attaches a seven-day weather forecast to every listing by contacting a weather service. Buyers can place an order for a listing. For orders above 500 kg, the buyer may start a price negotiation with the farmer before confirming. Every confirmed order requires an advance payment through an escrow payment service, and the money is released to the farmer only after the buyer confirms receipt.

Supermarket chains are registered as corporate buyers. They can do everything a normal buyer can, and they can also sign seasonal supply contracts with farmers. Agriculture extension officers can publish crop advisories and answer questions posted by farmers. A quality inspector inspects the crops for orders marked “quality checked” and uploads an inspection certificate. Transport partners accept delivery jobs and update the delivery status. The platform administrator resolves disputes between buyers and farmers.

Design a use case diagram based on the above scenario.

Question 4 CampusHub clubs and events

CampusHub manages clubs and events at a private university. All students log in with their university email, and the login is verified by the university's single sign-on service. Students can browse clubs, request club membership and register for events. Some events are paid; registering for a paid event also requires payment through the university payment portal. If an event is full, the student may join a waiting list instead.

Club executives are students who hold a post in a club. Besides everything a student can do, they can create event proposals, publish announcements and approve membership requests. Every event proposal must be approved by the club's faculty advisor before it is published. While creating a proposal, an executive may also request a venue booking, which is forwarded to the facilities office.

On the event day, volunteers scan each attendee's QR ticket at the entrance, and every scan validates the ticket against the registration list. After an event, attendees can give feedback, and the system compiles an event report that the executives and the advisor can view. The Office of Co-curricular Activities can suspend a club and view participation statistics for all clubs. Sponsors can view published events and submit sponsorship offers, which club executives may accept or decline.

Design a use case diagram based on the above scenario.

Question 5 RoktoBondhu blood donation network

RoktoBondhu is a blood donation network that links donors, patients and hospitals. Anyone can view the list of blood banks and read the donation guidelines. To take part, a person registers as a user and logs in. A registered user can post a blood request by giving the blood group, number of bags, hospital and deadline. If the request is marked as urgent, it is also broadcast by SMS to all eligible donors within 10 km.

A user who wants to donate can register as a donor. Donors can do everything a user can, and they can also accept blood requests. Whenever a donor accepts a request, the system checks the donor's eligibility, because a donor must wait at least 90 days between two donations. Donors can view their donation history and download a donation certificate.

Hospital blood banks update their stock of each blood group every day and confirm donations after collecting blood; confirming a donation always updates the donor's last donation date. Volunteers verify each new blood request by calling the hospital before the request is made public. All text messages are sent through an SMS gateway. The system administrator approves hospital accounts and blocks users who post fake requests.

Design a use case diagram based on the above scenario.

Part B · Activity diagrams

Questions 6–10. Show the flow from the initial node to the final node with actions, decisions with guards, merges, forks and joins where needed.

Question 6 Undergraduate admission

A private university handles undergraduate admission online. The applicant first fills in the online application form and pays the application fee. If the payment fails, the applicant may try again; after three failed attempts the application is cancelled. After a successful payment, the system generates an admit card, and on the test day the applicant sits for the admission test.

When the results are published, the process ends for an applicant who has not passed. An applicant who has passed is called for a viva. If the applicant passes the viva and seats remain, a seat is offered; if no seats remain, the applicant is placed on a waiting list and is checked again every time a selected candidate fails to confirm.

Once a seat is offered, two things happen at the same time: the applicant submits academic documents for verification, and the applicant completes a medical check-up at the university clinic. Only when both are cleared does the applicant pay the admission fee; if either one is not cleared, the offer is withdrawn. Finally, the system creates a student ID and the applicant collects the ID card.

Design an activity diagram based on the above scenario.

Question 7 Restaurant kitchen orders

A busy restaurant uses a kitchen display system for dine-in orders. A waiter takes the order on a tablet and sends it to the kitchen. The system checks every item against the inventory. If any item is unavailable, the waiter informs the customer, who either chooses a replacement or removes the item, and the order is checked again.

When all items are available, the head chef splits the order between three stations that work at the same time: the grill station prepares the main dishes, the cold station prepares salads, and the beverage station prepares drinks. Drinks are chilled before serving unless the customer asked for no ice. When all three stations are ready, the head chef performs a quality check. A dish that fails the check is sent back to its station to be remade and is checked again.

After the quality check is passed, the waiter serves the food. When the customer asks for the bill, the cashier prints it. The customer pays by cash or card, and card payments are verified by the card terminal. If the customer leaves feedback, it is recorded before the table is marked as free.

Design an activity diagram based on the above scenario.

Question 8 Small business loan approval

A bank processes small business loans through four parties: the applicant, the loan officer, the credit scoring system and the branch manager. The applicant submits a loan application with a trade licence and six months of bank statements. The loan officer checks whether all documents are attached; if any are missing, the application is returned and the applicant resubmits it with the missing documents.

When the documents are complete, two checks run at the same time: the credit scoring system calculates a credit score, and the loan officer visits the business to verify its address. When both checks are finished, an application with a credit score below 600 is rejected. Otherwise, a loan of up to Tk 5 lakh is approved by the loan officer, while a larger loan is forwarded to the branch manager. The branch manager either approves it, rejects it, or asks the applicant to reduce the amount; a reduced application goes back to the loan officer for the amount check.

Once a loan is approved, the loan officer prepares the agreement, the applicant signs it and the money is disbursed to the applicant's account. A rejected applicant is informed by email with the reason.

Design an activity diagram with swimlanes for the applicant, the loan officer, the credit scoring system and the branch manager.

Question 9 Garment export order

A garment factory in Gazipur handles export orders as follows. When a buyer sends a purchase order, the merchandiser prepares a sample. The buyer reviews the sample; if the buyer asks for changes, the sample is revised and sent again, and this repeats until the sample is approved.

After approval, the procurement team orders the fabric and the accessories (buttons, zippers and labels) at the same time. Accessories are always bought locally, but fabric is imported if the required fabric is not available locally. When both the fabric and the accessories have arrived, cutting, sewing and finishing are done one after another.

Next, the quality control team inspects a random batch. If the defect rate is above 2.5%, the whole lot is reworked and inspected again. When the lot passes, packing begins, and while the goods are being packed, the commercial team prepares the shipping documents at the same time. If the buyer has chosen sea freight, the cartons are sent to Chattogram port; for air freight, they are sent to the airport. The process ends when the cartons are handed over to the freight forwarder.

Design an activity diagram based on the above scenario.

Question 10 Smart irrigation controller

A smart irrigation controller on a rice field runs on its own. When it is switched on, it first runs a self-test. If the self-test fails, it sends a fault alert to the farmer's phone and shuts down. Otherwise, it starts its working cycle.

In each cycle, the controller takes three readings at the same time: the soil moisture, the water level in the field, and the rainfall forecast from the internet. If the internet is unavailable, the forecast is treated as “no rain”. After all three readings are taken, the controller decides what to do. If rain is expected within six hours, it skips irrigation for this cycle. If the soil moisture is below 30% or the water level is under 2 cm, it opens the valve and starts the pump. Otherwise, it does nothing.

While the pump runs, the controller checks the water level every minute and stops the pump when the level reaches 5 cm. If the pump has run for 40 minutes without reaching 5 cm, the controller stops the pump and sends a “possible leak” alert. At the end of every cycle the readings are logged. The controller then waits 30 minutes and starts the next cycle, unless the farmer has switched it off.

Design an activity diagram with a single activity final node.

Part C · Sequence diagrams

Questions 11–15. Show the lifelines, messages and returns in time order, with combined fragments, object creation and destruction where needed.

Question 11 Food delivery order

In a food delivery app, the Customer selects items from a restaurant and places an order through the MobileApp. The MobileApp sends the order to the OrderManager, which creates a new Order object and asks the PaymentGateway to charge the customer. If the payment fails, the OrderManager deletes the Order and the MobileApp shows an error to the Customer.

If the payment succeeds, the OrderManager sends the order to the Restaurant, which confirms it and returns the preparation time. The OrderManager then asks the RiderAllocator to find a rider. The RiderAllocator searches for available riders near the restaurant; if none is found, it widens the search radius by 1 km and searches again, up to a maximum of 5 km. If no rider is found even at 5 km, the OrderManager asks the PaymentGateway to refund the customer and cancels the order.

When a Rider accepts the job, the RiderAllocator returns the rider's details to the OrderManager, and the MobileApp shows them to the Customer. While on the way, the Rider sends its location to the OrderManager every 30 seconds. After delivery, the Rider marks the order as delivered, the OrderManager updates the Order's status and archives it, and the Order object is destroyed. Optionally, the Customer can rate the rider.

Draw the sequence diagram for the above scenario.

Question 12 Library self-checkout kiosk

A university library has a self-checkout Kiosk. A Member taps a library card on the Kiosk, and the Kiosk sends the card number to the MemberService, which checks the member's status. If the member has unpaid fines over Tk 200, the Kiosk shows the fine, asks the member to pay at the desk, and the session ends.

Otherwise, the Kiosk creates a CheckoutSession object. For each book the member places on the scanner, the Kiosk reads its RFID tag and asks the Catalogue whether the book can be borrowed. Reference books are rejected with a message. For any other book, the CheckoutSession creates a Loan object with a due date 14 days later, and the Catalogue marks the book as borrowed. A member can borrow at most five books; once the limit is reached, the Kiosk stops accepting books.

When the member presses Finish, the CheckoutSession asks the ReceiptPrinter to print a receipt; if the member chose an emailed receipt instead, the CheckoutSession asks the EmailService to send it. Finally, the Kiosk ends the session and the CheckoutSession object is destroyed.

Draw the sequence diagram for the above scenario.

Question 13 Intercity bus ticket booking

An intercity bus company sells tickets online. The Passenger searches for buses by route and date on the BookingPortal, which fetches the available trips from the TripService. The Passenger selects a trip and chooses seats, and the BookingPortal asks the SeatManager to hold them. If any chosen seat has already been taken, the SeatManager returns the list of unavailable seats and the Passenger chooses again; this repeats until all chosen seats are held.

When the seats are held, the SeatManager creates a SeatHold object that keeps them for 10 minutes. The Passenger then chooses a payment method: card payments are sent to the CardGateway, and mobile wallet payments are sent to the WalletGateway. If the payment is not completed within 10 minutes, the SeatHold releases the seats and is destroyed.

After a successful payment, the BookingPortal asks the TicketService to issue the ticket. The TicketService generates a unique PNR by itself and returns the e-ticket. If the Passenger has given a phone number, the TicketService also sends the ticket by SMS through the SMSGateway. Finally, the SeatManager turns the hold into a confirmed booking and the SeatHold object is destroyed.

Draw the sequence diagram for the above scenario.

Question 14 Smart home security alert

In a smart home security system, a MotionSensor that detects movement sends an alert to the HubController. If the system is disarmed, the HubController only records the event in its own log. If the system is armed, the HubController asks the Camera to start recording, and the Camera returns a live-stream link. The HubController then creates an IncidentReport object and sends a notification with the link to the OwnerApp.

The HubController resends the notification every 30 seconds until the Owner responds, for up to two minutes. If the Owner marks the event as a false alarm, the HubController asks the Camera to stop recording and closes the report. If the Owner marks it as a real threat, or does not respond within two minutes, the HubController turns on the Siren and asks the EmergencyService to send help to the home address.

In every armed case, at the end, the IncidentReport uploads the recorded video to the CloudStorage, which returns a link to the file. The HubController sends this link to the OwnerApp, and the IncidentReport object is destroyed.

Draw the sequence diagram for the above scenario.

Question 15 Assignment submission with plagiarism check

Students submit assignments through a university learning management system. The Student logs in to the LMSPortal, which checks the credentials with the AuthService. The Student selects an assignment and uploads a file. The LMSPortal sends the file to the SubmissionManager, which first asks the FileValidator to check it. If the file is not a PDF or is larger than 20 MB, the Student is asked to upload again; the Student can try at most three times.

When the file is valid, the SubmissionManager stores it in the FileStorage, which returns a file ID. If the deadline has passed, the SubmissionManager calculates a late penalty by itself and records it. Then the SubmissionManager creates a PlagiarismChecker object, which compares the file with the stored submissions one at a time until all have been checked and returns a similarity score. The PlagiarismChecker is then destroyed.

If the similarity is above 30%, the SubmissionManager flags the submission and the NotificationService emails the Instructor; otherwise, the submission is marked as received. In both cases, the LMSPortal shows the submission status and the similarity score to the Student.

Draw the sequence diagram for the above scenario.